
A Phone Samples Space, Not Time

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Abstract

OwnTracks reports on a timer or a distance, whichever fires first, so a moving phone sends one fix every D metres rather than every I seconds. We check $\min(I, D/v)$ on three iPhones reporting to ThingsBoard on EC2 (20,201 fixes, 312 device-hours), including one set to $D = 200$ m, whose step landed at 204 m. Turned around, the law gives speed from arrival times alone at 15.0% median error, and labels stop, walk and drive at 90.2%, 94.0% and 89.2%.

1 The law

OwnTracks in Move mode sends a report when either `locatorInterval` I or `locatorDisplacement` D is reached, so the gap between reports is

$$g(v) = \min\left(I, \frac{D}{v}\right), \quad v^* = D/I, \quad (1)$$

which with $I = 60$ s and $D = 50$ m switches at $v^* = 3$ km/h. Figure 1 shows what the phones recorded and Figures 2–5 show the two branches.

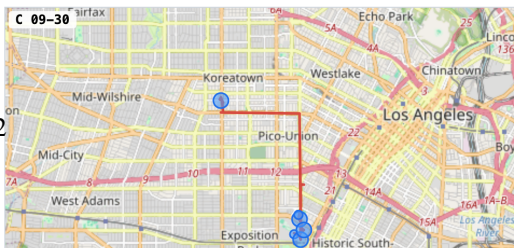
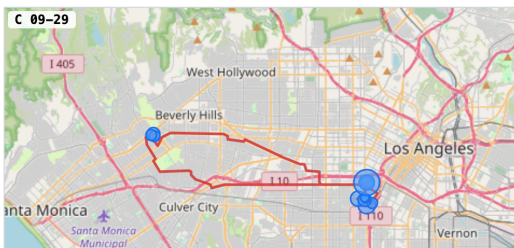
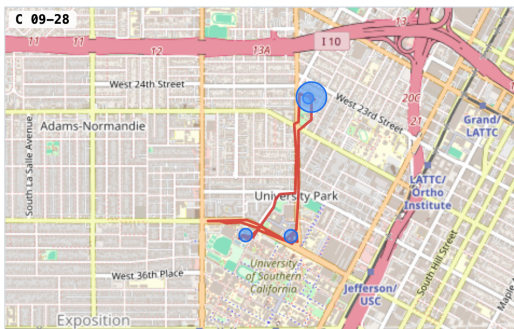
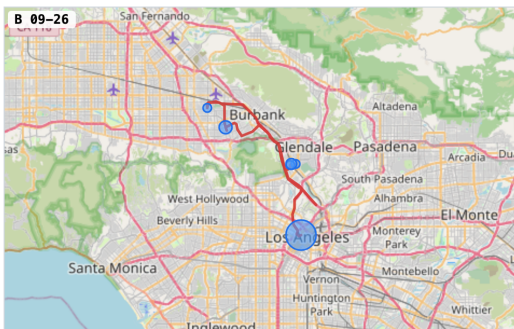
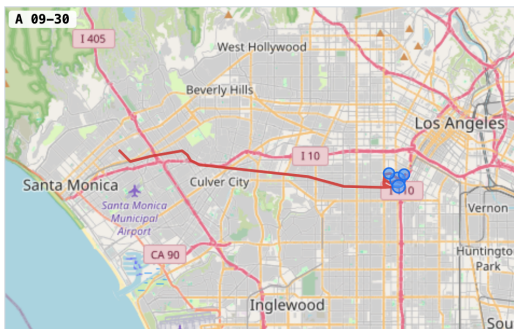
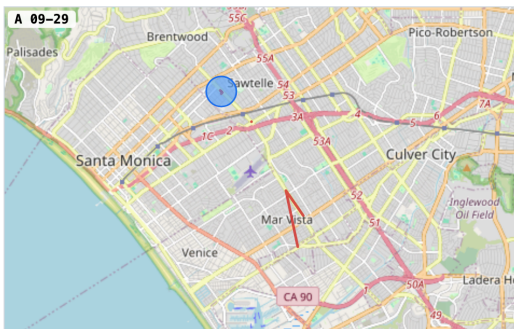
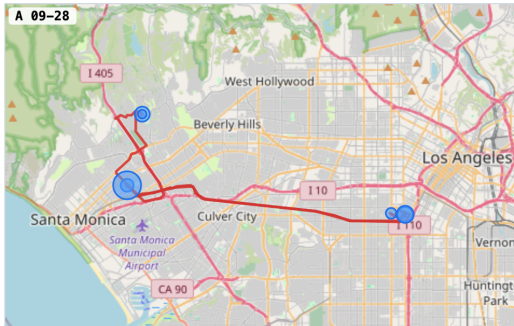
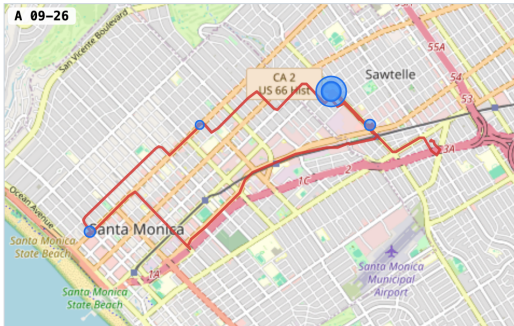
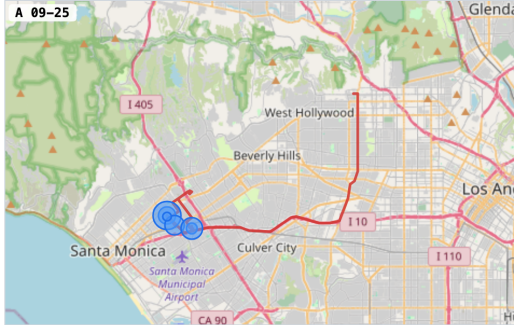
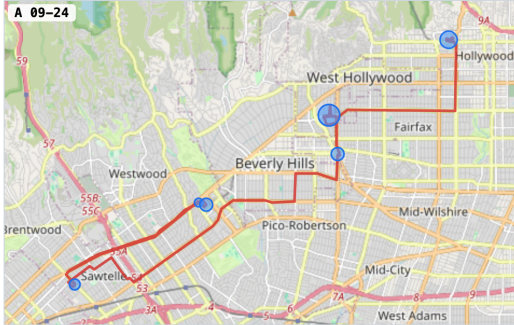
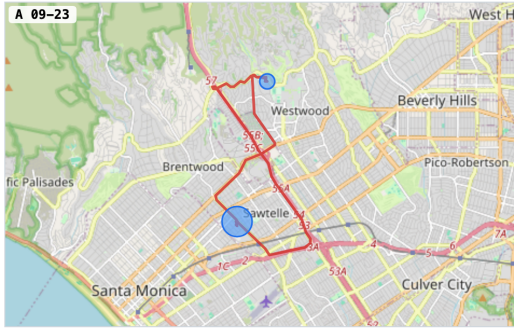
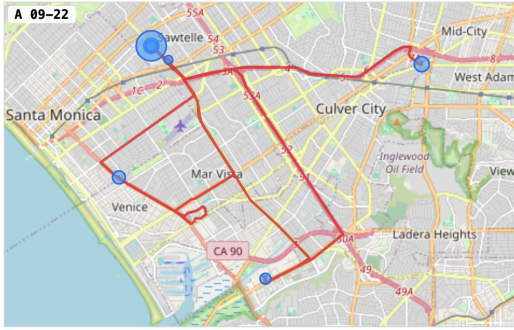
- The measured gap stays within a factor of 1.25 of Equation 1 on the two 50 m phones, using the app’s settings rather than a fit.
- With $D = 200$ m the step moves to 204 m and the crossover to 12 km/h, as predicted.
- The packet rate $\max(1/I, v/D)$ ranges from 60 reports per hour at rest to 2,000 at 100 km/h.

Adaptive GPS sampling [1, 2], transport-mode detection [3, 4] and map matching [5] all work from coordinates; like traffic analysis [6], we use only the timing.

2 Inverting it

For gaps shorter than I , speed is $\hat{v}_i = D/(t_i - t_{i-1})$, without reading the payload. Gaps equal to I only say the phone was slower than v^* . We compare against the same estimator using the actual distance between fixes, $\hat{v}_i^c = s_i/(t_i - t_{i-1})$. For segmentation, a gap at I is a **stop**, and shorter gaps split at 12 km/h into **walk** and **drive**. Table 1 and Figures 6–13 give the results.

Setup. Three iPhones running OwnTracks 26.2.3, one device token each, sending over HTTPS to ThingsBoard CE on EC2 from 2026-09-22 to 09-30. Two phones used $D = 50$ m and one $D = 200$ m. Gaps over 300 s (app suspended, Figure 15) are excluded, and grading uses $v \geq 5$ km/h.



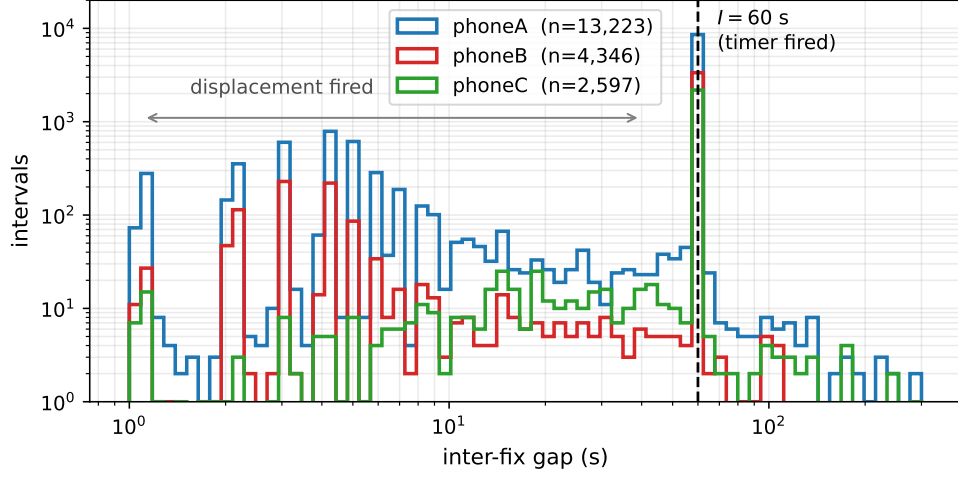


Figure 2: All inter-fix gaps: a spike at the timer and a tail below it from the distance trigger.

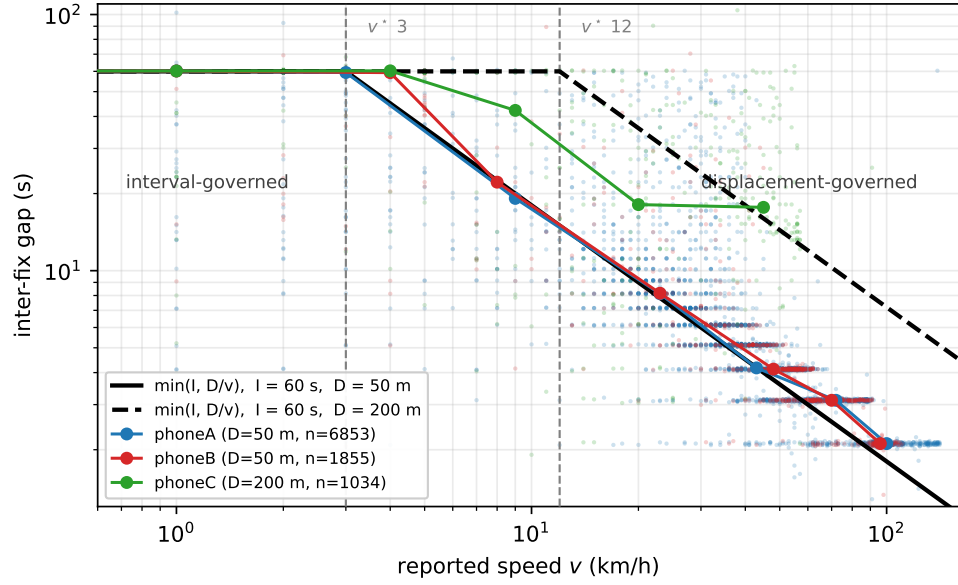


Figure 3: Gap against reported speed. The line is Equation 1 with the configured I and D , not a fit.

Table 1: Median relative error against reported speed, and segmentation accuracy.

	coordinates read	timing only, configured D	timing only, D_{eff}	3-class accuracy
phoneA	6.1%	17.5%	15.0%	90.2% ($n = 6,853$)
phoneB	5.9%	18.1%	13.3%	94.0% ($n = 1,855$)
phoneC ($D = 200$ m)	18.4%	59.2%	—	89.2% ($n = 1,035$)

Limitations. Only two values of D and one of I were tested, and phoneC follows the law only above 30 km/h. Ground truth is the phone’s own reported speed. All three phones are iPhones on one app version. The rendezvous detector (Figure 12) found one meeting, phoneA and phoneC on 09-30, so it shows the detector works rather than how often people meet.

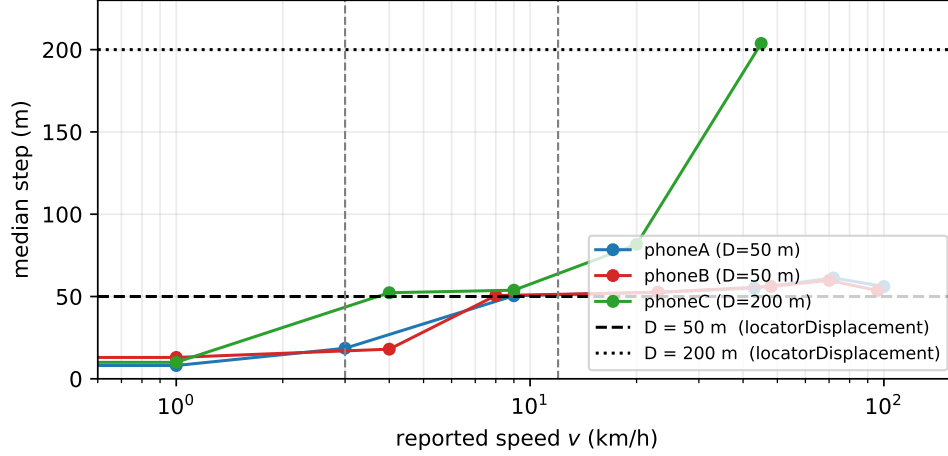


Figure 4: Distance between fixes. Above v^* the step stays near D : 52.6 and 54.0 m on the 50 m phones and 204 m on phoneC.

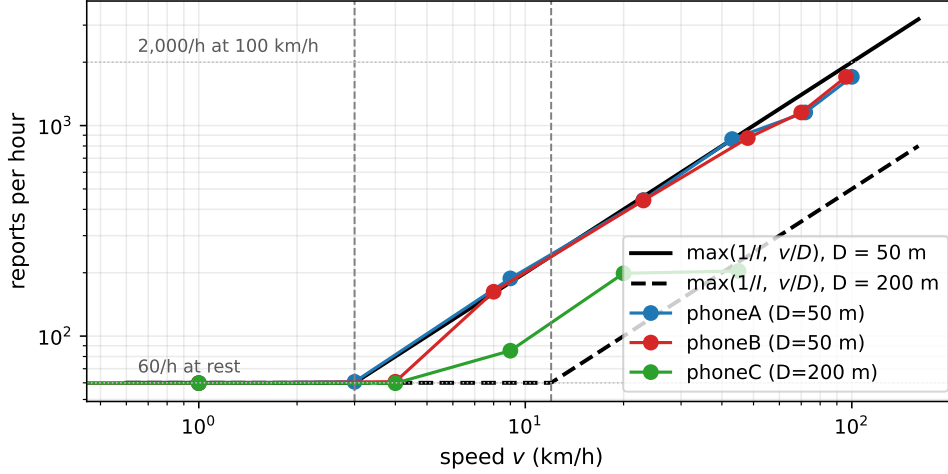


Figure 5: Reports per hour per device, $\max(1/I, v/D)$.

References

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- [2] J. Paek, J. Kim, and R. Govindan. Energy-efficient rate-adaptive GPS-based positioning for smartphones. In *MobiSys*, pages 299–314, 2010.
- [3] Y. Zheng, Q. Li, Y. Chen, X. Xie, and W.-Y. Ma. Understanding mobility based on GPS data. In *UbiComp*, pages 312–321, 2008.
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- [5] P. Newson and J. Krumm. Hidden Markov map matching through noise and sparseness. In *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '09)*, pages 336–343, 2009.
- [6] C. V. Wright, L. Ballard, S. E. Coull, F. Monrose, and G. M. Masson. Spot me if you can: uncovering spoken phrases in encrypted VoIP conversations. In *IEEE Symposium on Security and Privacy*, pages 35–49, 2008.

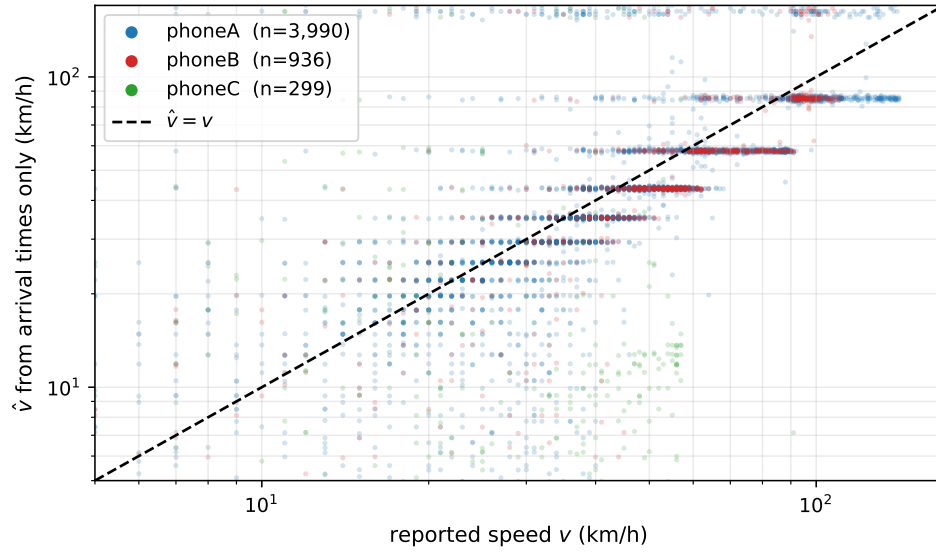


Figure 6: Speed from arrival times against the phone's reported speed.

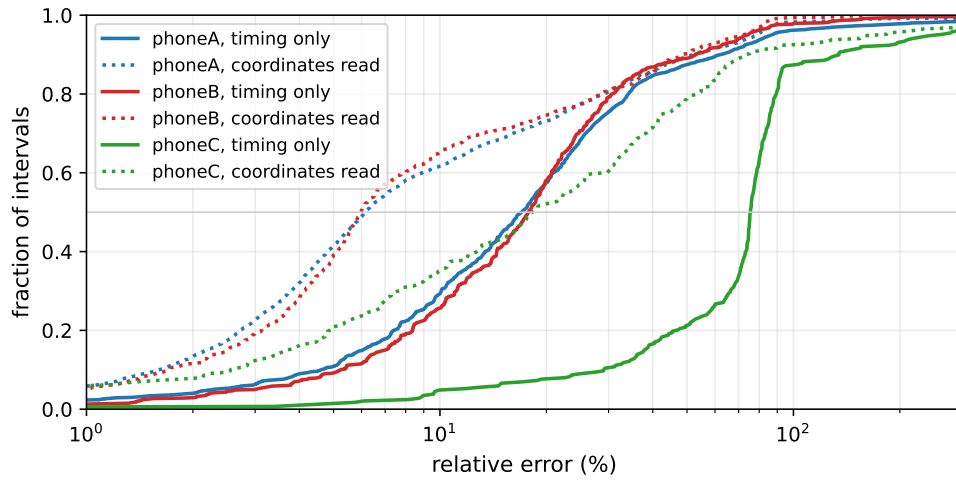


Figure 7: Error distribution, timing only against coordinates.

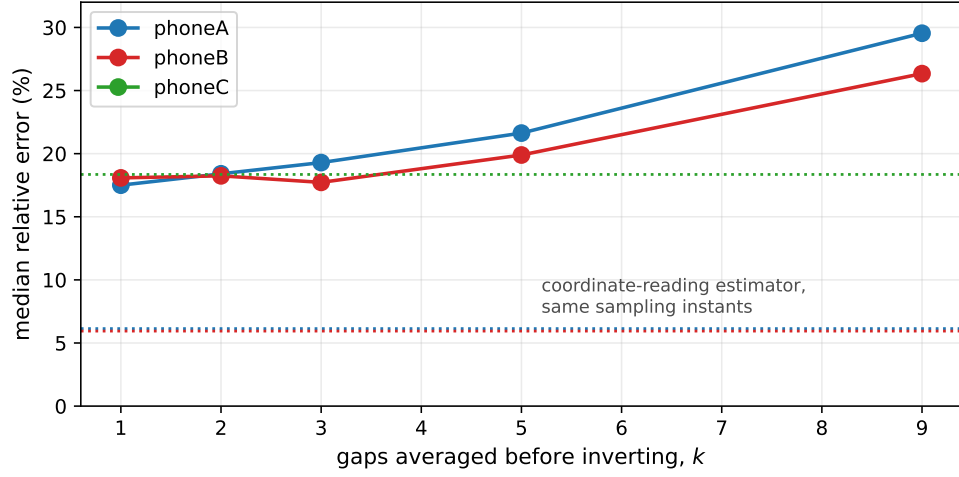


Figure 8: Averaging consecutive gaps makes the estimate worse.

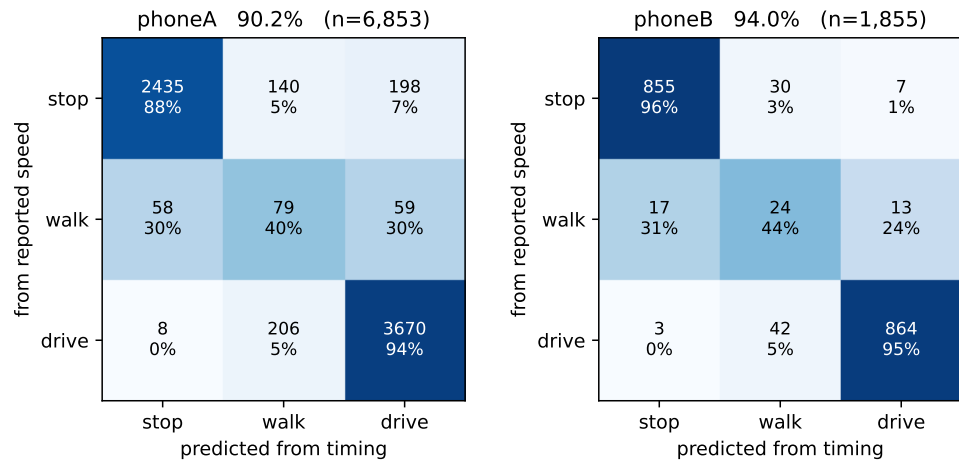


Figure 9: Segmentation confusion matrix. Most errors are in walk.

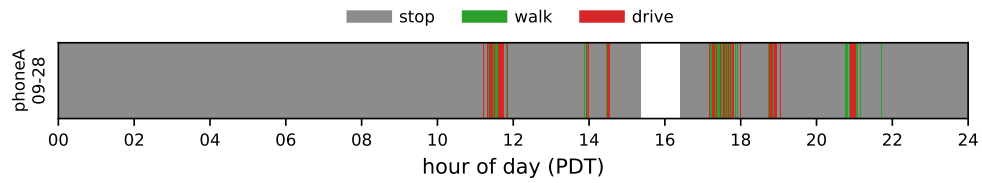


Figure 10: One day segmented from arrival times alone.

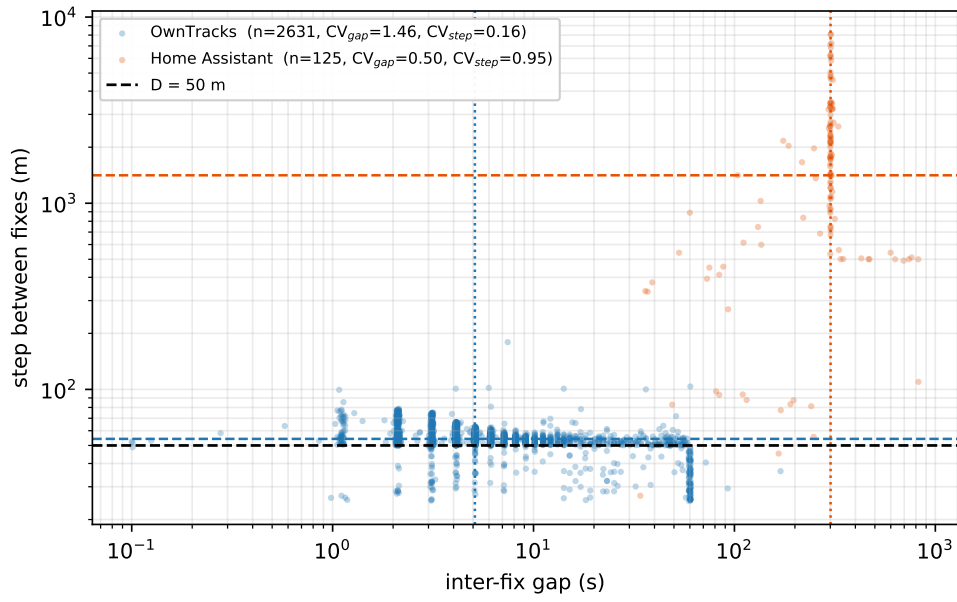


Figure 11: The same iPhone through two apps: OwnTracks fixes the step, Home Assistant fixes the gap.

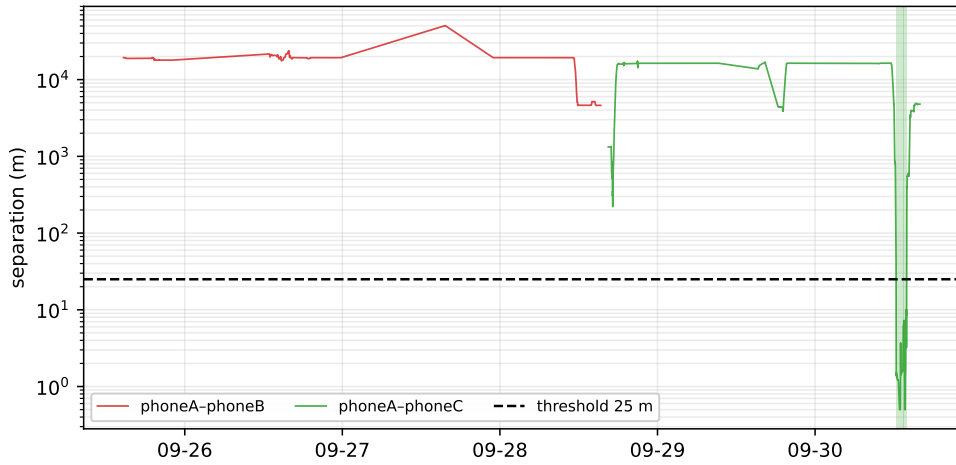


Figure 12: Distance between each pair of phones over time. Shaded spans are when two phones were together.

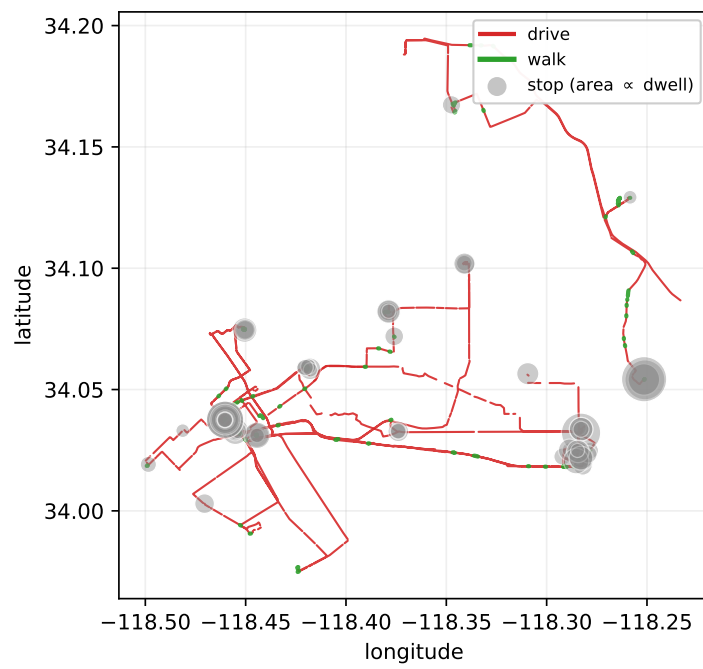


Figure 13: All tracks, classified from timing only. Circle area is time stopped.

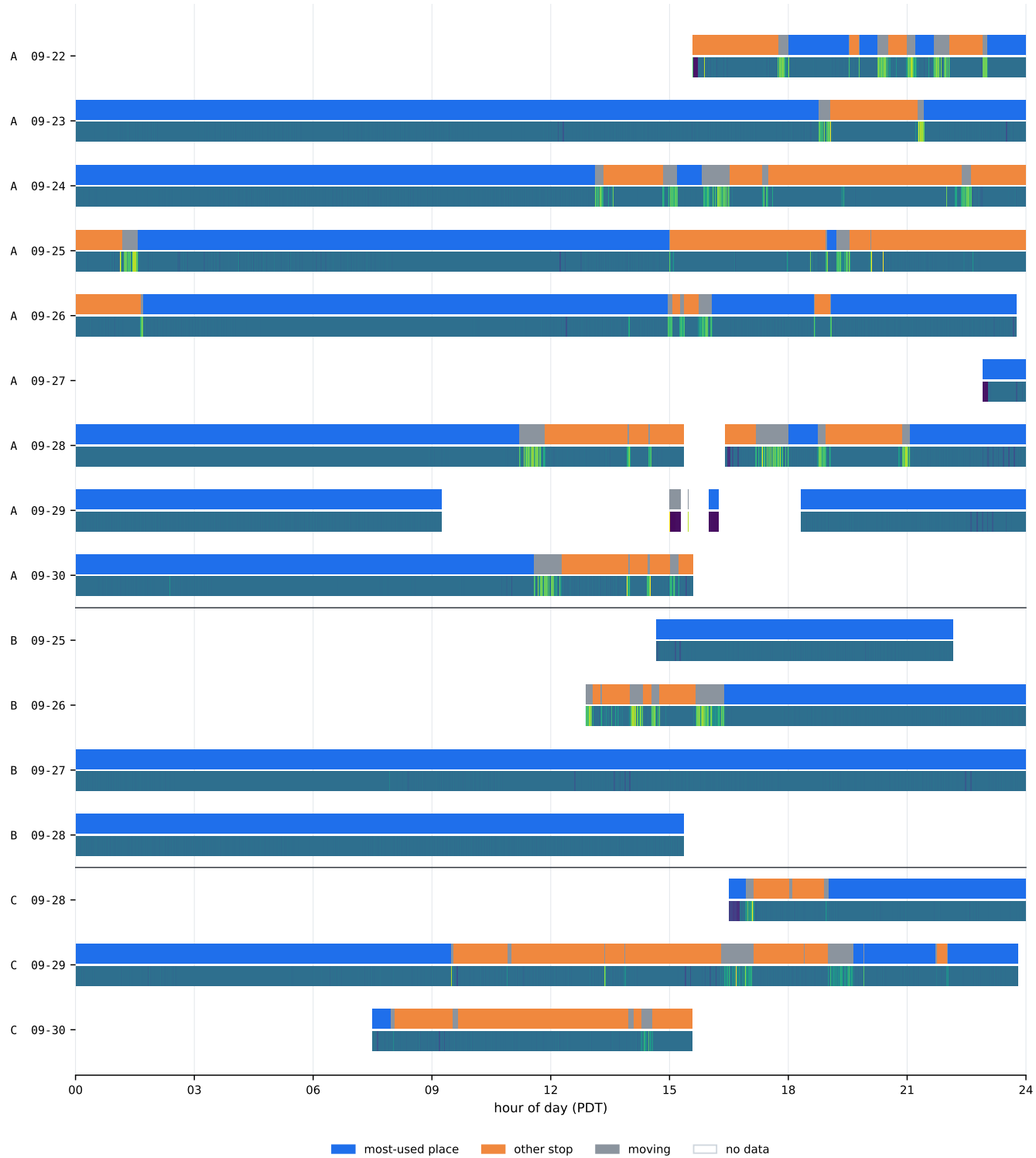


Figure 14: Every phone-day: where the phone was (upper band) and the gap between fixes (lower band).

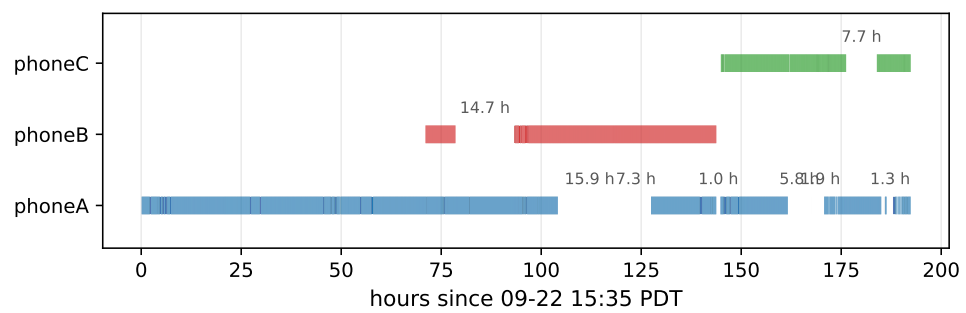


Figure 15: When each phone was reporting. The 15.9 h gap is iOS Low Power Mode.